

PopVax Robotics

Take-Home Assessment: Webcam-to-Bimanual Retargeting

Objective

Build a system that uses a laptop webcam to drive a bimanual OpenArm in MuJoCo. The operator stands in front of the camera and the arms follow their upper-body motion in real time.

We are evaluating how accurately human motion transfers onto the robot. There is no manipulation task - the quality of the tracking and the retargeting is the whole assessment.

Scene

Setting up the simulation is part of the task. You will need a bimanual OpenArm in MuJoCo with working actuators, sensible collision geometry, and joint limits that match the real arm. The OpenArm project publishes URDF/XACRO descriptions for single and dual-arm setups along with MuJoCo documentation - getting from there to a scene you can control at rate is yours to work out.

System Requirement

Precision is the point of this assessment. We are interested in how faithfully human motion maps onto a 7-DoF arm pair, not in getting something roughly waving on screen.

Your pipeline must expose an explicit intermediate retargeting target - the desired robot state, in the robot base frame, in robot units - sitting between perception and control. It should be logged every timestep and invariant to the operator's body size.

Choose your perception stack for accuracy. Lightweight landmark pipelines are usually not sufficient for precise retargeting; if you use one, justify it against the alternatives. You may train, fine-tune, or calibrate components if you think it helps.

The arms must be driven through actuators with physics stepping, not by writing joint positions into simulation state.

Constraints

Deadline: 28 July, end of day.

Sensing: one monocular RGB webcam. No depth cameras, VR trackers, multi-camera rigs, or markers.

Compute: a single consumer machine. A GPU may be assumed - state what you developed on and what the system needs to run.

Live Evaluation

We will run your system live, on our hardware, in front of our webcam, and move around in front of it ourselves. Your submission must include setup instructions precise enough for this to work without your involvement:

- environment setup, dependencies, and model weights, with versions pinned
- how to launch the system, and what the operator should see when it is running
- any calibration or startup procedure - where to stand, what the camera should see, how the system is engaged

Assume the person running it has not read your code.

Evaluation Criteria

- **Retargeting accuracy** - how faithfully the arm configuration reflects the operator's pose, including elbow placement and not just end-effector position
- **Tracking accuracy** - Cartesian error between the achieved end-effector pose and the commanded target
- **Latency** - end-to-end, camera to command applied
- **Framerate** - both perception and control, sustained under load

We expect the full pipeline to work end to end, live.

Deliverables

1. Reproducible code, the MuJoCo scene, and setup instructions.
2. Video showing the webcam feed and simulation side by side, across a range of motion - including fast movement, arms crossing the midline, and near the edges of the arms' reach.
3. **Design document (max 3 pages)**. Go through the stack component by component - scene construction, perception, retargeting, IK and redundancy resolution, filtering, threading and rate matching. For each:
 - what you built and why
 - what you considered instead and what it would have cost
 - how you would improve it, given more time or better sensing
4. Also cover: how the intermediate target is defined, how embodiment mismatch and monocular depth ambiguity are handled, and what breaks first as motion gets faster.

The improvement section matters. We want to see that you know what a better version of each component looks like, not just that you got one working.